

a bleak picture of our country, where we are ranked at 129 amongst 166 countries, and faring worse than countries such as Fiji, Maldives, Mongolia, Kazakhstan, and Kenya, among others. As far as Asian countries go, India's Internet penetration is only better than countries like Pakistan, Afghanistan, Iran, Laos, Nepal, and Bangladesh, which is again quite abysmal to say the least.

A lot of grass root level efforts are needed to ensure India's digital fortunes swing rapidly towards the better. This is something that the government will have to look at very seriously indeed, and sooner rather than later.

DEVICES

One of the biggest challenges that India faces is not just on the infrastructure side (challenges such as broadband network, cellular data coverage, etc) but also on the commodities side. In a digital economy, devices are pretty much the commodities, and we don't seem to have enough of them. Not only does India suffer from a scarcity of devices (wait, what?) but we also have a shortage of the right devices for the right people.

According to an Intel report on "Devices to Data Centres," India remains an underpenetrated country from a hardware perspective, too. Only 17 percent of the country has access to a PC, while the access to smartphone and tablet devices are pathetic down to less than 10 percent and 1 percent, respectively. Clearly, for a country of over a billion people, there's plenty of opportunity for hardware manufacturers to grow and make money. But they need to learn to marry the right devices with the right demographics.

Digital India can succeed only if Make in India propels manufacturers to make smartphones, PCs, and other interactive electronic gadgets that are relevant for local people. Because most of the things that we buy currently, when it comes to consumer electronics, gadgets and devices are manufactured and built according to global norms – not specifically to Indian norms. This is where a big change needs to take place if Digital India is to succeed, allowing manufacturers an opportunity to uniquely tailor a device's look, feel, and contents, depending on whether it's going to be sold and used in Uttar Pradesh or Tamil Nadu.

DIGITAL LITERACY

It's safe to say that India isn't a proud PC-literate country, far from it. But experts agree that we are a mobile literate country, due to the telecom boom of the last decade. However, being a mobile literate populace doesn't necessarily mean everyone knows the ins and outs of operating a smartphone and how to use Internet services off it. Let me explain.

In the same way that India missed the gaming console wave due to gross practicality (they were just too expensive over here and piracy was a great inexpensive substitute), we remain a poorly connected nation in terms of wired broadband. However, our wireless broadband infrastructure is certainly a lot better (despite being inadequate) than our wired connections. This obviously means that people have working knowledge of how to operate a phone and use it to make calls or send



messages – we see this behaviour across socio-economic stratas, right from our illiterate househelp to homeless people. Almost everyone is mobile phone savvy in our country. But does that mean they're at home to accessing the Internet with aplomb off a smartphone? Not quite.

This is where a major push is needed in improving the country's digital literacy efforts. And these need to be centred around the smartphone. Ensuring people who need access to Internet – villagers who need crop related update, or business folk who need on-the-second updates – get it in a way that they understand. At the same time, the push should be towards making the citizen Internet and tech-savvy.

DATA SECURITY

Let's be honest, the government of India's vision for Digital India is a massive one. We mean access to E-Healthcare portal which would not only cover online medical consultation but also hold online medical records of patients for convenient access. Similarly, E-Locker initiative to allow people to store all their sensitive government related documents securely online and share it across state and central agencies at the click of a button sounds like a great way to get out of bureaucratic quagmires going forward. The envisioned E-Sign capability to allow citizens to digitally sign documents by virtue of their AADHAR information is another great aspiration. Furthermore, there's also a road-map to integrate different government services and platforms, e.g. UIDAI, Payment Gateway, Mobile Platform, Electronic Data, and others.

All this will result in the government holding a lot of our sensitive data in the cloud. And if there's one thing we know about the cloud is that it's anything but secure.

Data security will be top priority as all these services are rolled out and integrated within each other on the back-end. This isn't an easy feat for any third-party vendor (such as Kaspersky, Quickheal or Symantec, for e.g.) without a doubt. In security parlance, a chain is only as strong as its weakest link, so this is one facet of Digital India which is pretty much the foundation on which the entire network sort of stands strong. There is absolutely no margin for error as far as data security is concerned.

LAST THOUGHTS

Whether it's education, healthcare, manufacturing, tourism, e-commerce or any other industry vertical, all of them can be made more efficient, optimum, far reaching and more profitable through the effective use of digital tools. We honestly believe that Digital India is a great initiative that has real transformative potential as far as India's fortunes are concerned. However, to realize the dream, a lot of work needs to be done and some key challenges need to be overcome. If we negotiate the road bumps smoothly and quickly, it will be only a matter of time before the dream of Digital India will be a reality that every Indian can savour. **■**